COMPx270: Randomised and Advanced Algorithms
Lecture 7: Nearest Neighbours and dimensionality reduction

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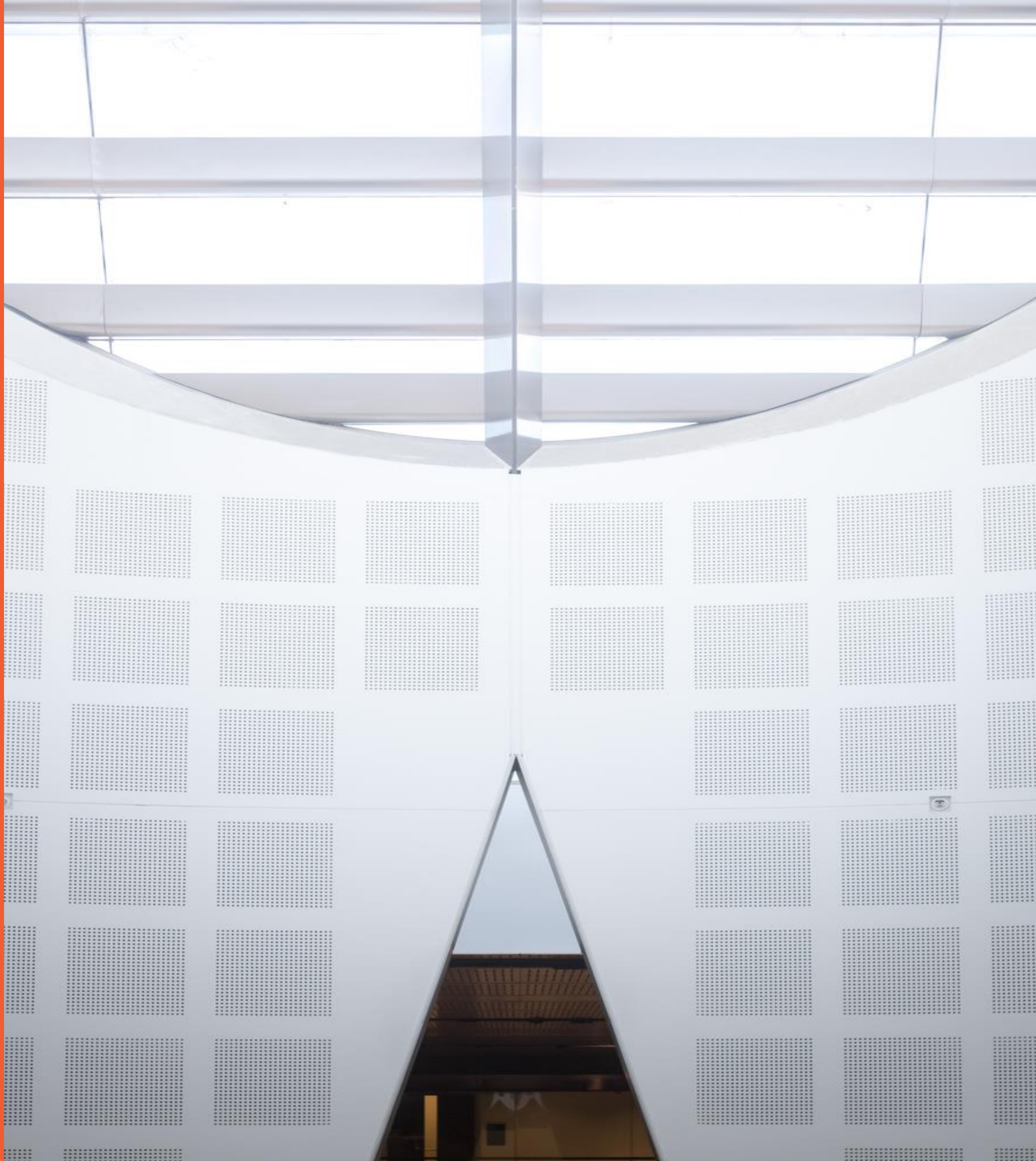


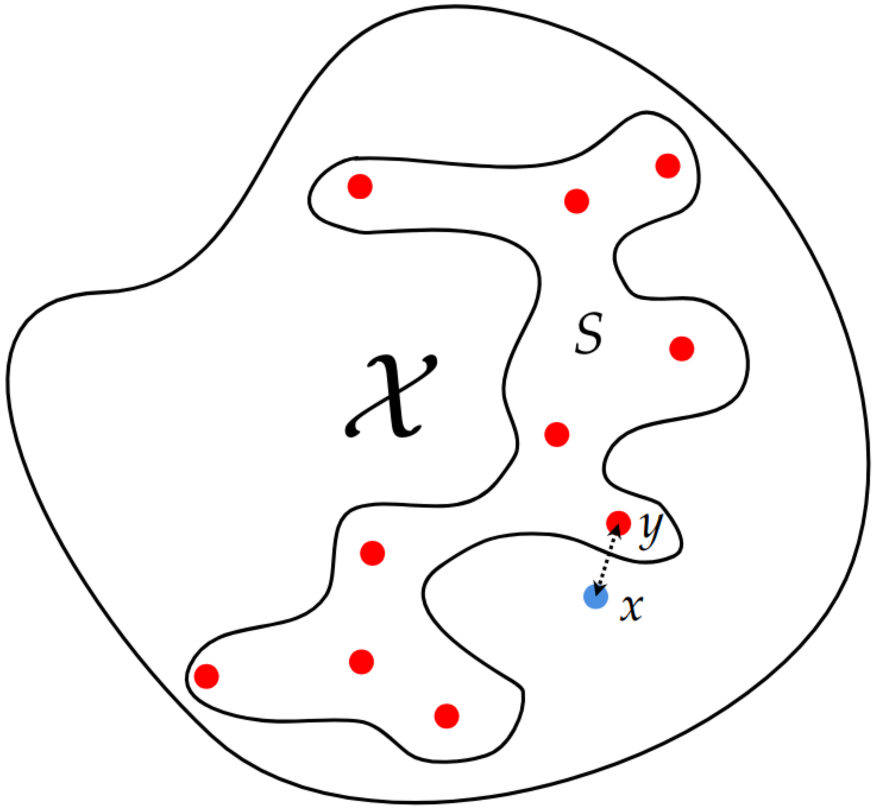
A question 🕷

You have n pictures, each 4096x4096 pixels, of venomous spiders. Someone finds a spider in their kitchen and sends you a photo, asking which type of spider it is and if it is venomous, **because they just have been bitten.**

How long will it take you?

Nearest Neighbour Search

$1 \ll d \ll n \ll 2^d$



Nearest Neighbour Search

$\textsc{Query}(x)$: on input $x \in \mathcal{X}$, return $y \in S$ such that

$$y = \operatorname*{argmin}_{y' \in S} \operatorname{dist}(x, y')$$

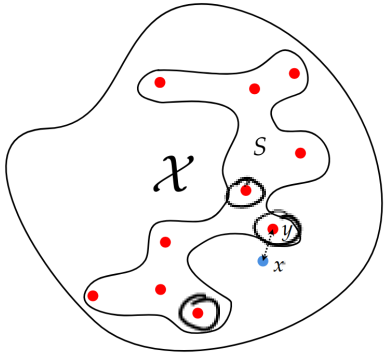
$1 \ll d \ll n \ll 2^d$

Lists? Voronoi? K-d trees? Hash tables?

	Query	Space
List		
Array* (for $\{0,1\}^d$)		
Voronoi		

Bad news...

Approximate Nearest Neighbour Search



QUERY(x): on input $x \in \mathcal{X}$, return $y \in S$ sort-of-minimising $\mathrm{dist}(x, y)$, that is, $\mathrm{dist}(x, y) \leq C \cdot \min_{y' \in S} \mathrm{dist}(x, y')$.

Dimensionality Reduction: the JL Lemma (Euclidean space)

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JL Lemma and ANN

QUERY(x): on input $x \in \mathcal{X}$, return $y \in S$ sort-of-minimising $\operatorname{dist}(x, y)$, that is, $\operatorname{dist}(x, y) \leq C \cdot \min_{y' \in S} \operatorname{dist}(x, y')$.

Beyond JL Lemma: Hashing!

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Locality-Sensitive Hashing

Definition 36.1. Let $0 \leq q < p \leq 1$, $r > 0$, $C > 1$, and $(\mathcal{X}, \mathrm{dist})$ be a metric space. Then a family of functions $\mathcal{H}$ from $\mathcal{X}$ to $\mathcal{Y}$ is a (r, C, p, q)-*Locality Sensitive Hash family* (LSH) if, for every $x, x' \in \mathcal{X}$,

- If $\mathrm{dist}(x, x') \leq r$, then $\Pr_{h \sim \mathcal{H}}[\, h(x) = h(x') \,] \geq p$;
- If $\mathrm{dist}(x, x') \geq Cr$, then $\Pr_{h \sim \mathcal{H}}[\, h(x) = h(x') \,] \leq q$;

and we say $\rho := \frac{\log(1/p)}{\log(1/q)} < 1$ is the *sensitivity parameter* of $\mathcal{H}$.

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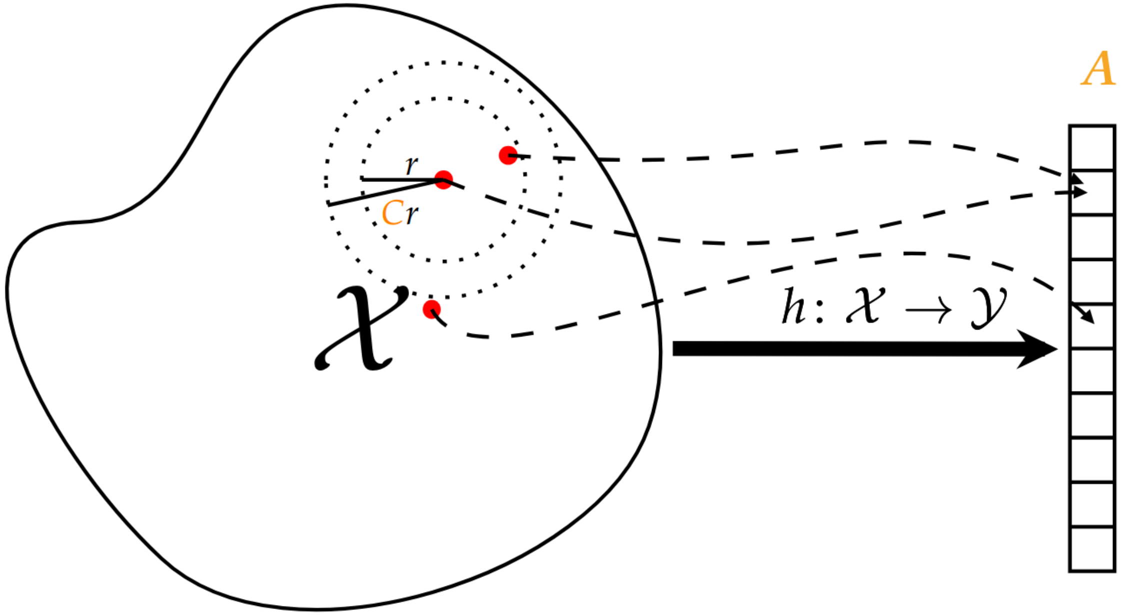

Locality-Sensitive Hashing: "Baby version"

$\textsc{Query}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\mathrm{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\textsc{Query}_r(x)$ returns an element $y \in S$ such that $\mathrm{dist}(x, y^*) \leq C \cdot r$;
- If $\mathrm{dist}(x, y) > C \cdot r$ for *every* $y \in S$, then, with probability 1, $\textsc{Query}_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

Locality-Sensitive Hashing: "Baby version" (1/6)

$\textsc{Query}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\mathrm{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\textsc{Query}_r(x)$ returns an element $y \in S$ such that $\mathrm{dist}(x, y^*) \leq C \cdot r$;
- If $\mathrm{dist}(x, y) > C \cdot r$ for *every* $y \in S$, then, with probability 1, $\textsc{Query}_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

Locality-Sensitive Hashing: "Baby version" (2/6)

$\textsc{Query}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\mathrm{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\textsc{Query}_r(x)$ returns an element $y \in S$ such that $\mathrm{dist}(x, y^*) \leq C \cdot r$;
- If $\mathrm{dist}(x, y) > C \cdot r$ for *every* $y \in S$, then, with probability 1, $\textsc{Query}_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

Locality-Sensitive Hashing: "Baby version" (3/6)

$\textsc{Query}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\mathrm{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\textsc{Query}_r(x)$ returns an element $y \in S$ such that $\mathrm{dist}(x, y^*) \leq C \cdot r$;
- If $\mathrm{dist}(x, y) > C \cdot r$ for *every* $y \in S$, then, with probability 1, $\textsc{Query}_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

$\textsc{Preprocess}(S)$:

```
for all x ∈ S do                    ▷ Insert all n elements in all k hash tables
    for all 1 ≤ t ≤ k do
        A_t[h_t(g_t(x))].INSERT(x)
```

$\textsc{Query}_r(x)$:

```
for all 1 ≤ t ≤ k do
    L_t ← A_t[h_t(g_t(x))]          ▷ "Candidates" in t-th hash table
    for all y ∈ L_t do
        if dist(x, y) ≤ C · r then
            return y                ▷ Found one!
return ⊥                            ▷ Did not find any.
```

Locality-Sensitive Hashing: "Baby version" (4/6)

$\textsc{Query}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\mathrm{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\textsc{Query}_r(x)$ returns an element $y \in S$ such that $\mathrm{dist}(x, y^*) \leq C \cdot r$;
- If $\mathrm{dist}(x, y) > C \cdot r$ for *every* $y \in S$, then, with probability 1, $\textsc{Query}_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

Locality-Sensitive Hashing: "Baby version" (6/6)

QUERY$_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\mathrm{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, QUERY$_r(x)$ returns an element $y \in S$ such that $\mathrm{dist}(x, y^*) \leq C \cdot r$;
- If $\mathrm{dist}(x, y) > C \cdot r$ for *every* $y \in S$, then, with probability 1, QUERY$_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

Theorem. *With suitable settings of k, ℓ and assuming $q, T = \Theta(1)$, the data structure for the "baby version of ANN" has space complexity $O(n^{1+\rho} d)$ and expected query complexity $O(n^\rho d)$.*

Locality-Sensitive Hashing: "Baby version" (👶)

Locality-Sensitive Hashing: "They grow up so fast" (🧑)

Locality-Sensitive Hashing: But... do they exist?

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